

Warm-up: Which one doesn't belong?

Poll Question

Each data table below represents a function. In your opinion, which one doesn't belong? (There may be multiple correct answers.) Discuss!

A

x	0	1	2	3	4
$f(x)$	0	2	4	6	8

B

x	-2	0	2	4	6
$g(x)$	6.1	6.4	6.7	7.0	7.3

C

x	0	1	2	3	4
$h(x)$	1	2	4	8	16

D

x	0	1	2	3	4
$\alpha(x)$	5	2	-1	-4	-7

Lecture 3 Linear Functions

Big Questions

- What are linear functions?
- Why are they useful?
- How can we recognize them?

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Linear Functions: Four Representations

Goal: After today's class you should be able to explain the characteristics of *linear functions* in terms of each of the four representations.

Tables

Graphs

Formulas

Words

Rate of Change

A local non-profit organization starts fund-raising for a new campaign, starting at the beginning of 2025. Every two months, they report their total funds raised toward their goal of \$49 000, shown in the table below.

Time t (months)	0	2	4	6	8
Funds y (thousands of \$CAD)	5	13	21	29	37

Poll Question

(a) At what **rate** are they raising money? How might you explain this **graphically**?

- A** 4 (\$CAD)
- B** 8 (per month)
- C** 8 (thousand \$CAD/month)
- D** 4000 (\$CAD/month)

Rate of Change (cont'd)

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Poll Question

(b) According to this data, when might you expect them to meet their fund-raising goal?

- A** 9 months
- B** 10 months
- C** 11 months
- D** 12 months

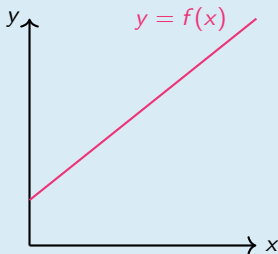
Slope

Definition

A **linear function** is a function whose graph is a straight line.

The **slope** of a linear function $y = f(x)$ can be calculated from the function values at two points x_1 and x_2 as:

$$\text{slope} = \frac{\text{rise}}{\text{run}} =$$



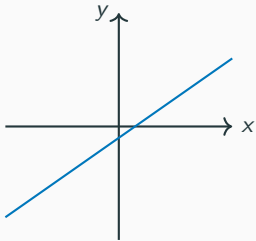
The slope of a linear function is sometimes called its “rate of change”.

Slopes of Lines

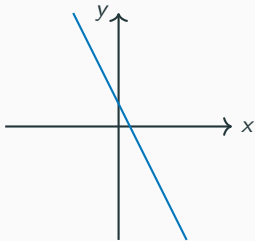
Poll Question

With your group, list the lines below in order of increasing slope. Assume that the axes of the graphs are identical. **Which line is first in your list?**

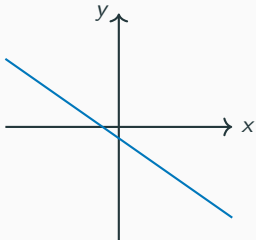
A



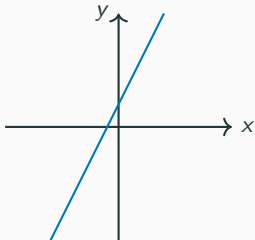
B



C



D



Formula for a Linear Function

Definition

The formula for a linear function can be put in the **slope-intercept** form

$$f(x) = mx + b.$$

The graph of $y = f(x) = mx + b$ is a line such that:

- m is the , the rate of change of y with respect to x .
- b is the , the value of y when x is zero.

Modelling with a Linear Function

Afsaneh is renting a car but forgets exactly how the rental pricing works. She remembers seeing on the car rental advertisement that if she drives 20 km it costs \$43, and if she drives 100 km it costs \$55.

Discussion Question

- (a) Assuming the rental cost y (in \$) is a linear function of how far she drives x (in km), determine a formula $y = f(x)$ for how much it will cost Afsaneh to rent the car.
- (b) Using plain language, give an interpretation of the slope and y -intercept. (More space on next slide.)

Modelling with a Linear Function (cont'd)

Discussion Question

(b) Using plain language, give an interpretation of the slope and y -intercept.

In part (a) we found $y =$. The slope being means that...

The y -intercept being means that...

Linear Functions: Examples in Four Representations

Now we have some examples for our different representations!

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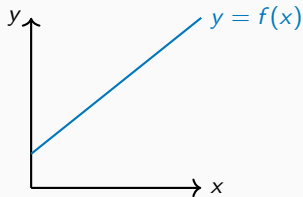
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Now we have some examples for our different representations!

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Graphs



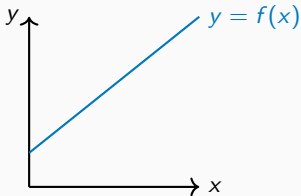
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Formulas

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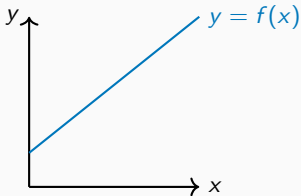
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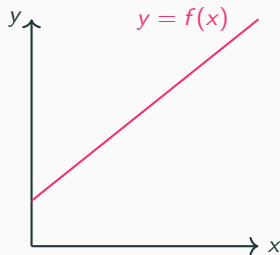
Words

Afsaneh is renting a car but forgets exactly how the rental pricing works. She remembers seeing on the car rental advertisement that if she drives 20 km it costs \$43, and if she drives 100 km it costs \$55.

Point-Slope Form

Definition

The **point-slope form** of a line with slope m passing through the point (x_0, y_0) is



After Class

If you know the y -intercept of the line is $y = b$, check that using $(x_0, y_0) = (0, b)$ in the point-slope form of the line will (after rearranging) yield the slope-intercept form!

Comparing the Forms of a Line

Discussion Question

Suppose a line has slope $m = -7$ and passes through the point $(3, 2)$.

- (a) Separately, find the **point-slope** and **slope-intercept** forms for the line.
- (b) Which form did you find was most efficient to find?

To do before next class:

- Complete today's Checkpoint!
- Please read Section 1.2 and finish Homework 1. (Homework 1 is currently available on WileyPlus and is due on Tuesday Sept 22 at 10pm.)
- Complete Week 1 Reflection Journal on Gradescope (due **tomorrow** at 10pm).

Drop-in hours are from 3-4pm on Wednesdays (OISE 5260) and Fridays (OISE 2286).

Have a great day!